Dynamic Modeling of a Data Center for Power System Stability Studies

Pam Paul Gyang, *Student Member, IEEE*, Pratyush Chakraborty, *Senior Member, IEEE*, Lasantha Meegahapola, *Senior Member, IEEE*, and Xinghuo Yu, *Fellow, IEEE*

***Abstract*—Data centers play a pivotal role in supporting digital transformation. However, they are among the most energy-intensive infrastructures as a result of increasing global computing load over the years. As data center power consumption continues to increase, examining their dynamic performance and interaction with the electrical grid is essential to enhance efficiency and ensure grid stability. This paper presents a comprehensive dynamic model of a 10 MW data center, incorporating 44,250 servers with a tandem hybrid heating ventilation and air-conditioning (HVAC) system to simulate real-world data center under varying workloads. The developed model is validated using experimental data that demonstrate high accuracy in predicting server power consumption, heat generation, and dynamic temperature. Furthermore, to verify the fidelity of the model to operate in dynamic power systems, the data center model is connected to WSCC 9-bus test system to analyze the impact of various grid disturbances on the data center operations, such as load variations and three-phase short-circuit faults. The model can be scaled up and is valuable for improving the accuracy of dynamic stability analysis of power systems under high integration of data center load. In addition, the model is proposed to provide a practical framework for assessing compliance with fault ride-through (FRT) requirements and enabling demand response participation.**

***Index Terms*—Data center, demand response, dynamic server model, fault ride-through, grid disturbance, stability studies.**

I. Introduction

A. Backgrounds

With the rapid expansion of cloud computing, Big Data analytics, the Internet of Things (IoT) and generative AI technologies, the reliance on data centers has increased considerably over the past decade. This increase has become a primary concern due to rising electricity demand, escalating operational costs, and environmental impact associated with energy-intensive data center operations [1]. According to [2], there were approximately 10,978 data centers worldwide as of December 2023, consuming around 2% of global electricity. This share is projected to rise significantly, reaching 4%–7% by 2026 [3]. A newly designed data center can consume enough electricity to power 37,000 homes annually, highlighting the urgency of managing energy efficiency [4]. It is also worth noting that since OpenAI's launch in 2022, AI adoption has surged, with a single query consuming 2.9 W-hours, nearly ten times that of a Google search (0.3 W-hours) [5], while training a GPT-3 model, which powers ChatGPT, requires up to 1.3 GWh of energy [6]. These statistics have prompted researchers in this domain to explore ways to reduce energy consumption in data centers, particularly as the growing dependence on renewable energy complicates the stability of the grid [7].

This growing energy demand has made data centers evolve beyond being static load facilities [8]. To address this issue, demand response programs enabling data centers to dynamically adjust their loads in response to grid disturbances have emerged as a promising solution [9]. However, determining the effectiveness of such a program largely depends on accurate modeling of data centers, considering their various subsystem dynamics. To highlight this, [10] shows that dynamic load analysis is more effective and offers greater detail for demand response studies compared to static models. A typical data center comprises of a server room (housing Information Technology equipment), a HVAC system, and power distribution equipment. Given that IT equipment and the cooling system are the most energy-intensive, accounting for approximately 60% and 30% of the total energy consumption of a data center, respectively [11], implementing demand-side flexibility on this equipment through demand response programs is essential for optimizing power consumption. Therefore, to effectively understand energy consumption patterns and operational flexibility within data centers and how it impacts the grid, a dynamic and precise model is required. In practice, system operators such as AESO (Alberta Electric System Operator) [12] have already emphasized that transmission-connected data centers require detailed time-series load profile modeling and validation of IT load and cooling load, to prevent negative impacts on grid stability. Specifically, data centers must comply with requirements for reactive power capability ($\pm$ 0.95 PF range) and ride-through performance for voltage and frequency disturbances. These requirements point out the pressing need for detailed phasor-domain models that represent server room load dynamics and motor-driven cooling systems.

Received 28 April 2025; revised 8 October 2025; accepted 23 November 2025. Date of publication 1 December 2025; date of current version 21 April 2026. Paper no. TPWRS-00746-2025. *(Corresponding author: Pratyush Chakraborty.)*

Pam Paul Gyang is with the Department of Electrical and Electronics Engineering, BITS Pilani Hyderabad Campus, Hyderabad 500078, India, and also with the School of Engineering, RMIT University, Melbourne, VIC 3000, Australia.

Pratyush Chakraborty is with the Department of Electrical and Electronics Engineering, BITS Pilani Hyderabad Campus, Hyderabad 500078, India (e-mail: pchakraborty@hyderabad.bits-pilani.ac.in).

Lasantha Meegahapola and Xinghuo Yu are with the School of Engineering, RMIT University, Melbourne, VIC 3000, Australia.

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TPWRS.2025.3639163.

Digital Object Identifier 10.1109/TPWRS.2025.3639163

0885-8950 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

B. Related Literature

Although various dynamic models have been developed for data centers, most emphasize individual components rather than a comprehensive system-wide approach. For example, studies by [13], [14] focus on real-time dynamic simulation of server power consumption, whereas [15], [16] focus on cooling system components. These models provide detailed insights, but they do not fully capture the overall power consumption dynamics of an entire data center. A different approach, demonstrated by [17], [18], involves mathematical modeling to calculate the total electrical power consumption of all internal components. While these models capture steady-state operations, they are unable to accurately predict dynamic behavior. To address the drawbacks of component-specific and steady-state models, [19] introduces a MATLAB/Simulink-based dynamic model that integrates energy consumption for IT and cooling devices, while [20] presents a dynamic model for data centers and implemented a power load emulator utilizing a hardware test bed. Nevertheless, these sophisticated models face limitations in accurately capturing servers' dynamic power consumption and the thermal loads generated by their operation, which are critical for sizing cooling system components. For instance, the model proposed in [20] assume that thermal conditions remain constant even if server workloads change within a time window. Additionally, these models often neglect environmental conditions, which have a substantial impact on the design and performance of modern data centers.

In analyzing the stability of power systems with data centers, [21] investigates the impact of transient disturbances on the load performance of the data center server. To better understand instability and resonance problems in large data centers, [22] and [23] present an impedance-based approach to analyze the low-frequency and resonance power stability issues in data centers, identifying the zero-sequence as the weakest link. Considering the impact of data centers' participation in demand response, [24] investigates the frameworks for data centers to provide dynamic power responses to support grid operations and improve grid reliability. Furthermore, data center load regulation has been examined in the context of stability of integrated power systems in [25], and the results show improved system stability using dynamic load shifting during off-peak periods. Research in [26] models critical components in data center power distribution systems for system simulations to study fault scenarios and protection requirements. Similarly, [20] proposes a dynamic model and converter-based emulator of a data center power distribution system to study its impact on grid stability. In addition, [27] recently proposed a dynamic model that explicitly incorporates UPS reconnection logics, induction motor-based cooling loads, and AI-driven pulsing server demands, demonstrating through case studies on the Irish transmission system that such features are essential to accurately capture FRT behavior, reconnection-induced instabilities, and periodic transients. Collectively, these studies contribute to improved design guidelines, specifications, and regulations aimed at enhancing the stability of power systems with high penetrations of data centers.

The current knowledge gaps identified from the existing literature, based on the research focus are summarized below.

1) Data centers' energy consumption and participation in demand response programs have made dynamic modeling of data centers practically useful. Therefore, it is necessary to conduct research on accurately representing the model's power consumption, thermal load, and temperature dynamics rather than just focusing on individual components. This is particularly important given the complex interdependencies between workload variations, cooling systems, and environmental conditions for sufficient control strategies that minimize power consumption.
2) Grid-connected data centers are now recognized as large-scale loads that can actively participate in grid operations. Hence, an accurate modeling approach is required to properly assess their impact. For linear models, the main disadvantage is that they fail to capture the nonlinear dynamics and complex interactions essential in such systems. This gap highlights the need for a system-level model in the phasor domain to support reliable power system studies.

C. Contributions

In this paper, the integrated electrical and thermal dynamic model of a 10 MW data center is developed and considered as an active load connected to the grid. The model captures both electrical power consumption and thermal interactions based on server utilization and environmental conditions, enabling a realistic representation of the data center's response to varying workload. Furthermore, the model, which incorporates FRT requirements, is connected to the grid for system stability studies in the phasor domain. Based on this, modeling and analysis approaches are proposed to support effective energy management strategies, enhance fault resilience, and facilitate the participation of data centers in demand response programs for grid stability. The main contributions of this study include the following.

1) The server model used in this study is developed and validated against experimental data to obtain precise power consumption, temperature, and heat generated. This modeling approach addresses the limitation of simplifying the heat generated to be the same as the power consumption [19], or alternatively, assuming a fixed temporal lag between them [20]. This validated model is then extended to develop the server room model by considering rack dimensions and individual server chassis, enabling a detailed representation of the thermal characteristics within the server room.
2) A comprehensive HVAC system model is developed using Simscape custom blocks in thermal liquid domain with high-level temperature control to coordinate the power consumption based on server workloads. This way, the total heat load generated from the server room is effectively managed and cooled in response to changing environmental conditions. Building on this, the electrical representation of the data center is modeled in Simscape

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

Electrical using variable load blocks and converters to capture its dynamic power demand.

3) We incorporated FRT requirements based on a system-level model developed in the phasor domain before connecting both the electrical and thermal subsystems of the data center into the grid. Thereafter, the stability of the integrated system is then evaluated under grid disturbances at the Point of Common Coupling (PCC) to validate the model's fidelity and suitability for dynamic and stability analyses in power system studies.
4) Finally, we proposed an integrated demand response framework that leverage both electrical and thermal flexibility of the data center to provide frequency support and enhance grid stability during load disturbances.

The remainder of the paper is organized as follows. In Section II, we propose a generalized model of a data center, along with a flow chart that illustrates its input and output variables. In Section III, we model the power characterization, heat load, and temperature dynamics of the server and validate them using experimental and simulation data. In section IV, we use the server model to develop a server room utilizing a tandem hybrid HVAC water cooling system. In Section V, we integrate the data center model into the WSCC-9 bus test system to analyze the impact of grid disturbances, FRT requirements and demand response participation. Section VI presents simulation results demonstrating the developed model's dynamic performance during a weekly workload and stability analysis, and then a baseline FRT and demand response capability. Finally, we conclude and discuss future work in Section VII.

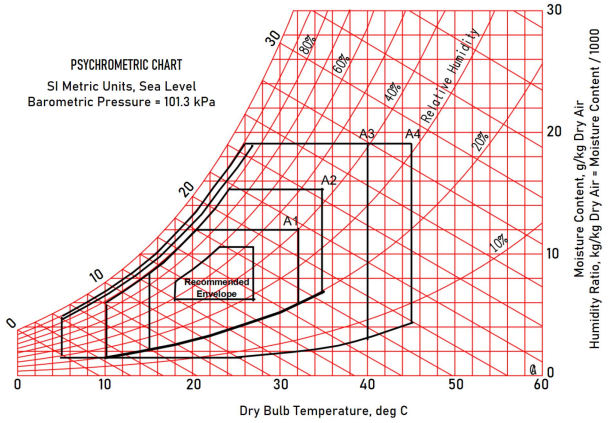


Fig. 1. ASHRAE data center environmental classes [28].

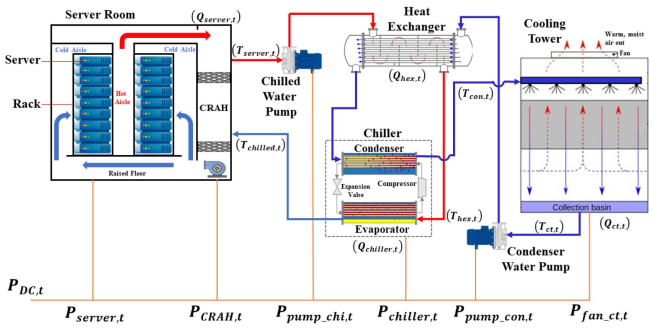


Fig. 2. Detailed model of the data center.

II. Data Center Model

A data center is fundamentally considered as a load facility, which consumes power primarily through its IT equipment and cooling infrastructure. Therefore, developing an integrated dynamic model of these two components should be prioritized when evaluating the impact of a data center load consumption on the electrical grid. As discussed in [8], detailed modeling of data center loads, including their dynamic and adjustable characteristics, is essential for capturing the complex dynamics and flexibility of data center power consumption. In this section, the overall data center model and its major components are introduced to develop the dynamic simulation model. The model is developed in the root mean square simulation domain.

A virtual model of a 10 MW data center with 44,250 servers arranged in a hot aisle containment configuration is developed, following ASHRAE Class 1 thermal guidelines, which specify an allowable temperature range of 18-27 °C and relative humidity of 20-80% for equipment reliability [28]. As described in [29] and [30], the hot aisle containment improves air supply efficiency leading to a more uniform temperature distribution and enhanced cooling performance, as shown in Fig. 1. The data center HVAC system system is modeled in MATLAB/Simscape using custom blocks [31] to manage the substantial heat generated by the servers. In a 10 MW facility, the servers produce approximately 10 MW of heat, which must be effectively removed from the server room and discharged into the external environment. Due to its scale and high thermal load, the data center adopts a cooling architecture distinct from smaller systems, relying on water as the primary cooling medium because of its high specific heat capacity. It is modeled within the thermo-liquid domain, as depicted in Fig. 2. The system consists of two separate water loops: a closed chilled water loop that absorbs and rejects heat ($Q_{server,t}$) at a temperature ($T_{server,t}$) through the Computer Room Air Handler (CRAH), which uses chilled water supplied at temperature ($T_{chilled,t}$). An open condenser water loop then expels heat ($Q_{ct,t}$) into the environment through a cooling tower. The cooling tower supplies water through the condenser water loop at a temperature ($T_{ct,t}$). The separation of these loops ensures clean circulation within the server room. Heat transfer between the two water loops is facilitated by a heat exchanger, characterized by a heat transfer rate ($Q_{hex,t}$) and an output temperature ($T_{hex,t}$), coupled with a chiller operating at a heat transfer rate ($Q_{chiller,t}$). The heat exchanger is a more cost-effective solution as it transfers heat without consuming power, whereas the chiller operates using a refrigeration cycle that requires energy. As a result, the chiller is activated and used more when the external environment is too hot for the heat exchanger alone to sufficiently cool the chilled water. The total power consumption of the data center, $P_{DC,t}$, comprises of the power consumed by the server room ($P_{server,t}$), the CRAH unit ($P_{CRAH,t}$), the chilled water pump

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

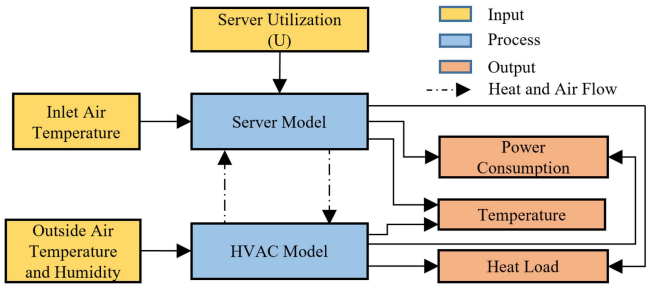


Fig. 3. Flowchart of the data center thermal and power consumption.

($P_{pump_chi,t}$), the chiller ($P_{\text{chiller},t}$), the condenser water pump ($P_{pump_con,t}$), and the cooling tower fan ($P_{fan_ct,t}$).

The flow diagram in Fig. 3 shows a process that begins with key input variables: server utilization and temperature, which are used to compute the load on the server, as well as environmental temperature and humidity, affecting the efficiency of the cooling system. The heat generated by the servers, based on server utilization, is transferred and regulated by the HVAC system. The final output variables include data center power consumption, heat load, and temperature.

III. Server Model

This section mathematically models the server's power consumption characteristics, heat load, and temperature dynamics. We then validate the model using experimental and simulation data, which is then used to develop a server room.

A. Server Power Characterisation

Modeling a server's dynamic behavior requires understanding the primary factors influencing power consumption, thermal load, and temperature dynamics. From a first-principles perspective, this modeling approach divides the power consumption into two main components: static power P_{static} and dynamic power $P_{dynamic}$, given in (1) [32]. Here, static power represents the baseline energy consumed when the server is idle, while dynamic power fluctuates based on workload, clock frequency, and supply voltage, making it a key driver of thermal behavior and temperature changes in the server [17].

$$P_{server} = P_{dynamic} + P_{static} \tag{1}$$

According to [33], dynamic power is widely regarded as the circuits' primary source of power dissipation. It arises within the server from three primary components: switching power, short-circuit power, and leakage power, as shown in (2)

$$P_{dynamic} = P_{switching} + P_{short_circuit} + P_{leakage} \tag{2}$$

Switching power $P_{switching}$ occurs during the charging and discharging of load capacitance as the server switches states with increasing clock frequency. The short-circuit power P_{short_circut} results from transient currents flowing through transistors during these transitions, which are also frequency-dependent. Leakage power $P_{leakage}$, however, is a baseline consumption from continuous leakage currents within transistors, independent of the server's state and influenced mainly by transistor properties and temperature.

$$\begin{cases} P_{switching} = t_{sw} \cdot \frac{1}{2} C V_{dd}^2 f \\ P_{short_circuit} = t_{sw} \cdot I_{sc} \cdot V_{dd} \cdot f \\ P_{leakage} = n_g \cdot I_{leakage} \cdot V_{dd} \end{cases} \tag{3}$$

Where C in (3) is the capacitance, V_{dd} is the supply voltage, I_{sc} is the short-circuit current, t_{sw} is the switching duration, which represents the fraction of transistors switching per cycle, f is the clock frequency, n_g is the number of gates in the circuit, and $I_{leakage}$ is the leakage current. The static power, which is directly proportional to the device count, is given by:

$$P_{static} \propto I_{static} \cdot V_{dd} \tag{4}$$

If β represents a constant that combines the effects of transistor count, design characteristics, and leakage current, which flows when the server is in an idle state with n transistors, then I_{static} can be represented as the leakage current. Thus, (4) can be reformulated as follows:

$$P_{static} = n \cdot I_{leakage} \cdot V_{dd} \tag{5}$$

and the total server power can be represented as:

$$P_{server} = \beta \cdot V_{dd} + t_{sw} \cdot I_{sc} \cdot V_{dd} \cdot f + t_{sw} \cdot \frac{1}{2} C V_{dd}^2 f \tag{6}$$

However, calculating the server power consumption using the above is challenging due to the influence of frequency, voltage, and other variables. Various models, from simple empirical methods to complex physics-based approaches, have been developed to estimate power consumption based on system characteristics and data over the years. The study by [11] categorized 47 server power models into types, including linear regression [34], power function [11], non-linear [34], and polynomial models [35]. The cubic polynomial model demonstrates a superior fit with minimal error, making it good enough to use for accurate power estimation across performance metrics. Using server utilization alone to estimate server power consumption is both efficient and accurate, as processors and memory are the primary contributors to power usage, while disk and network power remain nearly constant [19], [36]. This simplified approach also minimizes performance impact by eliminating the need to track multiple metrics, as in (6). Instead, the total server power consumption as a function of server utilization can be represented as:

$$P_{server} = a1 + a2 \cdot U_{server} + a3 \cdot U_{server}^2 + a4 \cdot U_{server}^3 \tag{7}$$

Where a_1 to a_4 are coefficients that can be determined using curve-fitting techniques applied to the average power consumption data across a range of server utilization levels.

B. Server Heat Load and Temperature Dynamics

This subsection focuses on developing the server thermal model to account for heat generation. It focuses on accurately modeling the temperature distribution, thermal resistance, and the effectiveness of thermal management throughout the system.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

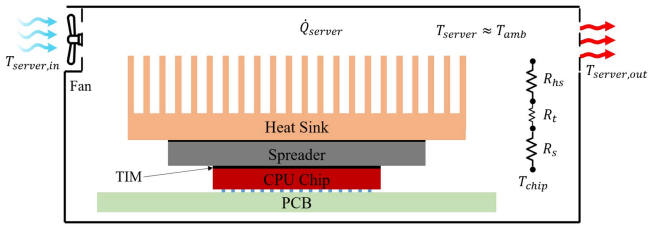


Fig. 4. Thermal management system for a server Model.

Fig. 4 illustrates a simplified thermal model for a server system, showing the path of heat transfer from the CPU chip to the external environment. The CPU generates heat that is conducted through a thermal interface material (TIM) layer, a spreader, and a heat sink [37]. The TIM layer, labeled as R_t, ensures efficient conductive heat transfer between the CPU chip and the spreader. The overall thermal resistance of the system, R_{total}, consists of the sum of the resistances from the heat sink R_{hs}, the TIM R_t, and the spreader R_s. Cool air enters the server through a fan at $T_{server,in}$ and absorbs heat from the system, eventually exiting at $T_{server,out}$. The heat dissipation rate is represented by $\dot{Q}_{server}$. The model is based on the first law of thermodynamics, which states that the rate of change in internal energy $\frac{dU}{dt}$ is determined by the balance between the power generated by the server, P_{server} and the heat dissipated $\dot{Q}_{server}$;

$$\frac{dU}{dt} = P_{server} - \dot{Q}_{server} \tag{8}$$

Assuming C_{server} is the server's thermal capacitance, the equation becomes:

$$\frac{dU}{dt} = C_{server} \cdot \frac{dT_{server}}{dt} = P_{server} - \dot{Q}_{server} \tag{9}$$

Based on the figure, the total thermal resistance, R_{total}, is given by the sum of resistances of each layer in the heat transfer path:

$$R_{total} = R_{hs} + R_t + R_s \tag{10}$$

where R_t is the thermal resistance of the TIM, which is critical in ensuring efficient conductive heat transfer between the CPU and the spreader. The heat dissipation rate $\dot{Q}_{server}$ can be expressed as:

$$\dot{Q}_{server} = \frac{T_{server} - T_{server,in}}{R_{total}} \tag{11}$$

Substituting $\dot{Q}_{server}$ into the energy balance equation, we get:

$$C_{server} \cdot \frac{dT_{server}}{dt} = P_{server} - \frac{T_{server} - T_{server,in}}{R_{total}} \tag{12}$$

Rearranging this equation provides a differential form that describes the rate of temperature change of the server:

$$\frac{dT_{server}}{dt} = \frac{P_{server} - \frac{T_{server} - T_{server,in}}{R_{total}}}{C_{server}} \tag{13}$$

Equation (13) can be used to predict the transient temperature response of the server based on the power input P_{server}, the inlet temperature $T_{server,in}$, the thermal resistances and capacitance of the system. To further understand the temperature dynamics in a server cooling system, we analyze the relationship between the heat generated by the server and the resulting temperature increase in the cooling fluid as it flows through the system. According to the fundamental principles of heat transfer, the heat carried away by the cooling fluid is given by:

$$\dot{Q} = \dot{m} \cdot c_p \cdot (T_{server,out} - T_{server,in}) \tag{14}$$

This equation represents the heat transfer rate $\dot{Q}$ within the fluid as it absorbs energy from the server's components. Here, $\dot{m}$ is the mass flow rate of the fluid, c_p is its specific heat capacity, while $T_{server,in}$ and $T_{server,out}$ are the inlet and outlet temperatures of the fluid, respectively. Assuming that all heat generated by the server is transferred to the cooling fluid, then (11) and (14) can be formulated to obtain $T_{server,out}$:

$$\frac{T_{server} - T_{server,in}}{R_{total}} = \dot{m} \cdot c_p \cdot (T_{server,out} - T_{server,in}) \tag{15}$$

if (15) is rearranged, then $T_{server,out}$ becomes;

$$T_{server,out} = \left(1 - \frac{1}{\dot{m} \cdot c_p \cdot R_{total}}\right) T_{server,in} + \frac{1}{\dot{m} \cdot c_p \cdot R_{total}} T_{server} \tag{16}$$

TABLE I
THERMAL DESIGN POWER (TDP) OF THE SERVER COMPONENTS

Device	Reference Designator	TDP (W)
Processors	CPU0	95
	CPU1	95
DIMMS	A0	0.81
	A3	0.81
	A6	0.81
	B0	0.81
	B3	0.81
	B6	0.81
Chipsets	ICH	4.3
	IOH	27.1
Fans	Fan1	0.7 - 5.64
	Fan2	0.7 - 5.64
	Fan3	0.7 - 5.64
	Fan4	0.7 - 5.64
Power Supply	PS	4.5% of server power
Hard Drives	HDD	7.8

C. Validation of the Server Model

The developed thermal model is validated by comparing its predictions against experimental and simulated data for server power consumption, temperature response, and heat dissipation reported in prior studies [14], [38], [39]. A server model based on Open Compute Project, featuring a dual-socket Intel Xeon 5500 or Xeon 5600 motherboard with 18 DIMM slots in a 1.5 U form factor configuration is considered. This server is equipped with two Intel CPUs, each with a thermal design power (TDP) of up to 95 W. Additionally, it includes an Intel 5500 I/O Hub chipset for efficient data handling. The TDP values for each of these components, listed in Table I, provide essential input

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

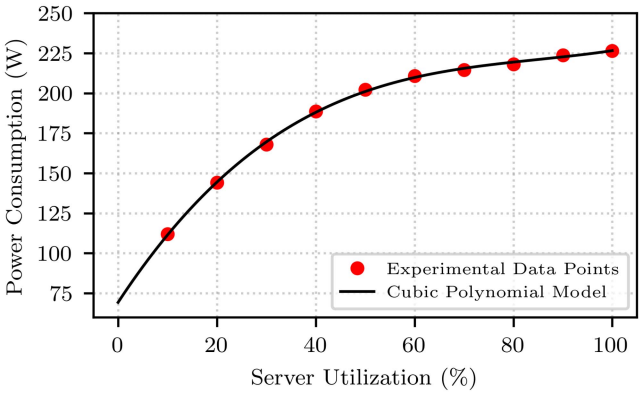


Fig. 5. Power vs server utilization with cubic polynomial fit.

TABLE II
EXPERIMENTAL DATA OF CPU UTILIZATION, POWER CONSUMPTION, FAN SPEED, AND AIRFLOW

Server Utilization (%)	$P_{server,m}$ (W)	RPM (RPM)	CFM (CFM)
10	111.92	1758	20.74
20	144.18	1790	21.32
30	167.97	2347	29.93
40	188.61	2673	33.95
50	202.27	2999	39.35
60	210.72	3062	40.40
70	214.52	3064	40.43
80	218.03	3482	48.64
90	223.86	3221	43.23
100	226.49	3608	51.68

TABLE III
PARAMETERS FOR SERVER THERMAL AND AIRFLOW MODEL

Parameter	Symbol	Value
Operating Server Temperature	T_{die}	68°C
Initial Server Temperature	T_0	18°C
Heat Capacity of Server	c_{server}	30 J/K [42]
Maximum Fan Speed	RPM_{max}	3608 RPM
Specific Heat Capacity of Air	C_p	1005 J/(kgK)
Maximum Airflow	CFM_{max}	51.58 CFM
Air Density at atp	ρ	1.225 kg/m^3

parameters for validating the model's predictions on temperature and heat dissipation.

The experimental data from sources [38], [39] were utilized to perform curve fitting for average power consumption across a server utilization range from 10% to 100%, in increments of 10%, at a constant ambient temperature of 30 °C. The server's fans were powered internally during testing to simulate operational conditions. The curve fitting, which models power consumption as a function of server utilization, is illustrated in Fig. 5. This approach yields (17), as derived from (7), with a fit accuracy of 0.9992. The data supporting this curve-fitting process are summarized in Table II.

$$P_{server,m} = 69.282 + 4.707 \cdot U_{server} - 0.0515 \cdot U_{server}^2 + 0.000202 \cdot U_{server}^3 \quad (17)$$

The server's thermal response and power consumption are modeled using specific parameters that characterize its thermal and airflow properties. Table III presents the values used, including the operating and initial server temperature, the specific heat capacity of air, the heat capacity of the server, and fan characteristics such as maximum Revolutions Per Minute (RPM) and Cubic Feet per Minute (CFM). To determine the thermal resistance (R_{total}) of the server system, (18) adapted from [40] and the reformulated data in [39] were used.

$$R_{total} = C_1 + \frac{C_2}{CFM^k} \quad (18)$$

In this model, $C_1 = 1$, $C_2 = 0.1911$, and $k = 1.5$, where C_1 and C_2 are constants related to the airflow properties and the CPU package, while k represents the turbulence level in the airflow [41]. These values were chosen based on experimental observations, providing an accurate fit to the thermal resistance curve across the entire fan speed range [14].

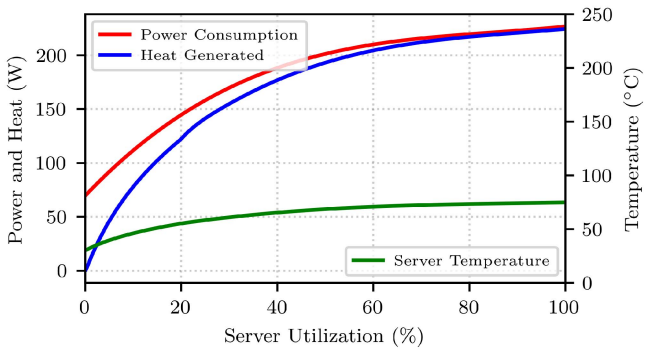


Fig. 6. Thermal and power response of server model.

The simulation results in Fig. 6 show a strong fit with experimental data, confirming the model's accuracy in predicting thermal behavior across server utilization levels. At full load, the model estimates a server temperature of 73.49 °C, closely matching the experimental result of 72.27 °C with a minor deviation of 1.69%, which is well within the acceptable range for thermal models (typically below 5%). The power consumption and heat generation curves exhibit a nonlinear rise with increasing server utilization, reflecting expected thermodynamic behavior in server systems. The temperature profile on the right y-axis follows an anticipated trend, gradually increasing with load. It is essential to model the heat generated ($\dot{Q}$) separately from power consumption (P), rather than assuming equivalence. While P is related to ($\dot{Q}$), they diverge in real scenarios as some power dissipates in other forms, and ($\dot{Q}$) often lags behind P due to thermal inertia, especially under dynamic loads. Capturing this delay is essential for accurate thermal response modeling, particularly in high-performance servers where thermal delays can affect peak temperatures and cooling requirements.

IV. SERVER ROOM WITH HVAC SYSTEM

In this section, we utilize the validated server model to develop the data center server room. Additionally, we provide the mathematical formulas necessary for sizing the HVAC system.

The model utilizes an optimized rack arrangement, with each 42 U rack assuming homogeneous servers, efficient airflow, and

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

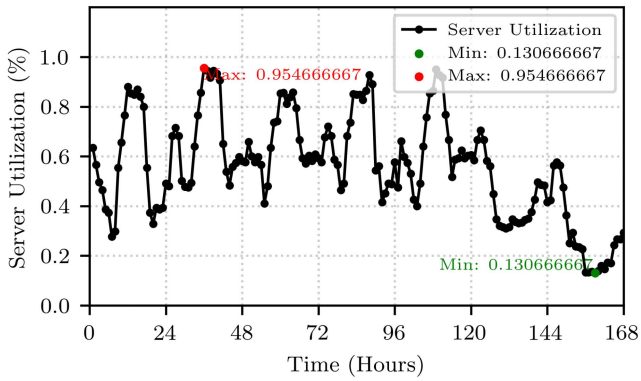


Fig. 7. Weekly data center workload on an hourly basis [14].

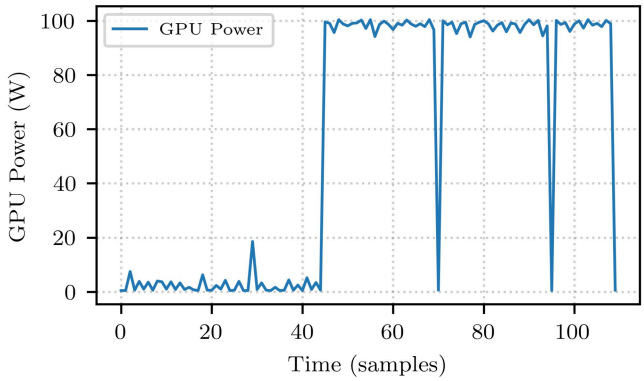


Fig. 8. GPU Power Consumption Over Time.

a heat isolation system, which maintains the thermal environment within recommended ASHRAE limits. By indexing each server as i and each rack as j, the model allows for detailed thermal and electrical dynamics parametrization, capturing complex interactions under variable loads. If M represents the total number of racks in the data center and N_j the number of servers in rack j, then the total power consumption and heat generated by all the racks j (W) is;

$$\begin{cases} P_{total} = \sum_{j=1}^{M} \sum_{i=1}^{N_j} P_{i,j} \\ \dot{Q}_{total} = \sum_{j=1}^{M} \sum_{i=1}^{N_j} \dot{Q}_{i,j} \end{cases} \tag{19}$$

therefore, the outlet temperature supplied for each rack j can be determined using the energy balance equation.

$$T_{out,j} = T_{in,j} + \frac{\sum_{i=1}^{N_j} \dot{Q}_{i,j}}{\dot{m}_{air,j} \cdot c_p} \tag{20}$$

Since thermal mass significantly impacts dynamic simulation [43], it is imperative to consider the impact of each server and rack. According to an experiment done by [44], a server's average specific heat capacity is $0.6\,\text{kJ/kg} \cdot \text{K}$, while the average mass of a server is 6 kg [43]. If the mass of an empty 42 U rack made of steel with a specific heat capacity of $0.49\,\text{kJ/kg} \cdot \text{K}$ is assumed to be 100 kg, then the total thermal mass can be calculated in J/K. Finally, the server workload in Fig. 7 is a weekly workload of a telco data center obtained from existing research work [14]. This workload, with a minimum and maximum server utilization of 0.13 and 0.95, respectively, will be used to model the dynamic power consumption of the data center. As shown in Fig. 8, GPU power consumption varies significantly over time, with several periods of high power usage close to 100 W GPU utilization [45]. This pattern further reflects the dynamic nature of server workloads in real-world data center operations.

The developed model in this section can also be adopted for heterogeneous data center, by establishing separate parameter sets for each cluster of servers based upon their respective thermal and power characteristics. The same set of underlying dynamic equations presented in this work can be employed for each server group while adjusting coefficients such as rated power, utilization power sensitivity, and thermal capacitance.

A. *Heating, Ventilation, and Air Conditioning Model*

The HVAC system employs a tandem hybrid cooling approach using indirect evaporative techniques [30]. The design responds to outdoor temperature and humidity; when the ambient wet bulb temperature exceeds 18 °C, the refrigeration chiller turns on, and both evaporative cooling and refrigeration operate together in a combined mode. The two main advantages of this approach are its energy efficiency and better heat management. Therefore, with components working in series, the system can optimize performance and reduce energy consumption. The tandem hybrid cooling system must be sized appropriately to ensure energy efficiency and effective cooling. All the system components must be carefully sized as well, to handle peak cooling loads for the data center during ambient temperatures. Firstly, the cooling tower capacity is sized according to the data center's maximum anticipated heat load when the system runs in combined mode [46]. Next, the chiller is sized and controlled to activate at a specific temperature threshold to provide backup cooling. At the same time, the pumps and heat exchanger are also chosen to ensure decent and continuous flow between components without excess pressure drops or energy losses. The mathematical equations provided in Appendix A, together with the modeling parameters specified in Table IV, are used to size and model the cooling system.

V. Grid-Connected Data Center

This section explores the dynamic interaction between the data center and the grid, focusing on electrical load modeling, FRT capability, and participation in the demand response program.

To evaluate the impact of power system dynamics on the proposed 10 MW data center, we integrated the model into the WSCC 9-bus test system. We began with a small-signal stability analysis to assess how the system responds to minor disturbances, which also helps validate the performance of the control system under normal operating conditions. This analysis focuses on the chiller controller as shown in Fig. 9. A proportional integral controller is used to ensure robust and interpretable control performance in regulating the cooling demand within the data center. Additionally, it offers predictable dynamics, minimal

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COOLING SYSTEM AND PIPING PARAMETERS

Parameter	Value
Cooling tower volumetric flow rate	32,292 L/min
Cooling tower hot-water inlet temperature	28 °C
Cooling tower cold-water outlet temperature	22 °C
Cooling fan power	95 kW
Cooling tower moist-air volumetric flow rate	463.38 m^3/s
Chiller power	700 kW
Chiller COP	10
CRAH power	460 kW
Chilled water pump capacity	21,300 L/min
Chilled water pump head	28 m
Chilled water pump brake power	127 kW
Condenser water pump capacity	32,292 L/min
Condenser water pump head	33 m
Condenser water pump brake power	217 kW
Heat exchanger port cross-sectional area	0.3 m^2
Heat exchanger mass flow rate	250 kg/s
Heat exchanger heat transfer rate	4000 kW
Chilled water loop mass flow rate	370 kg/s
Condenser water loop mass flow rate	536 kg/s
Server pipe diameter	0.02 m
Server pipe length	12 m
Server pipe thickness	0.002 m
Number of server pipes	1600
Pipe density ρ	7800 kg/m^3
Pipe specific heat c_p	500 J/(kg·K)
System port area	0.3 m^2

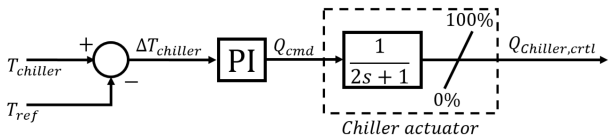


Fig. 9. Chiller control system.

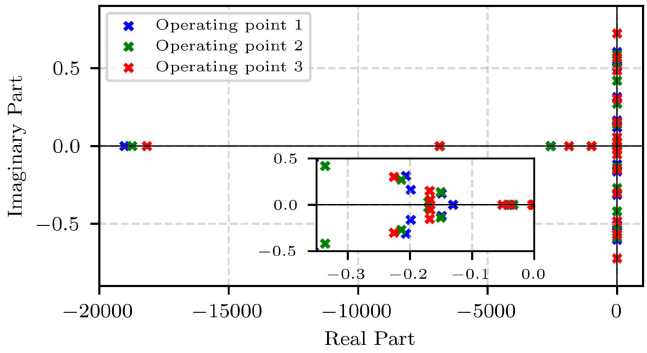


Fig. 10. Eigenvalue plot for the control loop of the chiller.

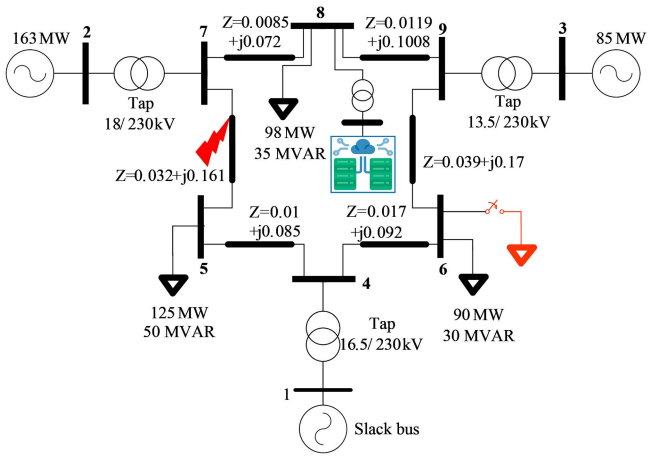


Fig. 11. Modified WSCC 9-bus test system with data center and case studies.

data dependency, and less tuning issue, which is well suited for the model's fidelity level. It interacts with other components to maintain the desired temperature while accounting for varying server utilization, which is essential to prevent thermal stress and optimize overall energy efficiency. By dynamically adjusting the cooling output in response to fluctuations in server utilization and environmental conditions, the controller enhances the data center's capacity to respond effectively to both internal and external disturbances, making it a critical component for achieving robust and resilient thermal management. Due to the complexity introduced by the thermal liquid process, Simulink's Linearization Toolbox is used to obtain the eigenvalue distribution of the chiller controller. This complexity arises from constraints such as flow continuity, pressure balance, and energy conservation within the HVAC system, which complicate the linearization process.

The linearization process has generated 27 states, capturing the dynamic interactions between the chiller, heat exchanger, cooling tower, and the server room's thermal mass. The eigenvalue analysis shown in Fig. 10 confirms that all eigenvalues have negative real parts at various operating points (t = 50, 100, and 10000 s), indicating that the system is stable. The linearized model provides meaningful insight into the small signal stability characteristics of the data center model.

Following this validation step, the data center's thermal and electrical model is integrated into PQ bus 8 of the WSCC 9-bus test system [47]. The test bench generators are equipped with a synchronous machine-type AC1C excitation system, featuring a lead-lag block to enhance the model dynamics associated with the voltage regulator and the PSS1A stabilizer. The droop coefficients of the generators are set to a maximum value of 5%. We employ MATLAB/Simulink in to analyze the impact of various disturbances on the data center, such as changes in load within a 5%, 15%, and 25% of the total system load (315 MW) to capture a representative range of small, medium, and large disturbance that could occur in practical grid scenarios, three-phase short-circuit fault, FRT capability and demand response participation as illustrated in Fig. 11.

A. Load Modeling

When considering a data center connected to the grid as an active load, its electrical power consumption at the PCC can be represented as follows:

$$S = P + jQ \tag{21}$$

where P and Q are the data center's total active and reactive power, respectively. To capture the dynamic load consumption of the data center when connected to the grid, we model the time series load demand for each major component using a variable PQ Simscape electrical load block. The server room, CRAH,

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

TABLE V
LOW- AND HIGH-VOLTAGE AND FREQUENCY RIDE-THROUGH REQUIREMENTS

Voltage Ride-Through Requirements	
RMS Voltage V (pu)	**Min. Ride-Through Time (s)**
$V > 1.25$	May ride through or trip
$1.20 < V \leq 1.25$	1
$1.10 < V \leq 1.20$	1800
$0.90 \leq V \leq 1.10$	Continuous
$0.75 \leq V < 0.90$	3
$0.65 \leq V < 0.75$	2
$0.45 \leq V < 0.65$	0.3
$V < 0.45$	0.15
Frequency Ride-Through Requirements	
Frequency F (Hz)	**Min. Ride-Through Time (s)**
$F > 61.8$	May ride through or trip
$61.2 < F \leq 61.8$	299
$58.8 \leq F \leq 61.2$	Continuous
$57 \leq F < 58.8$	299
$F < 57$	May ride through or trip

chiller, and cooling tower fan are modeled as PQ loads, each represented by a series R-L branch, with the per-phase resistance and inductance for component i calculated as (22).

$$R_i = \frac{P_i V_{rb}^2}{P_i^2 + Q_i^2}, \qquad L_i = \frac{Q_i V_{rb}^2}{2\pi f_{grid}\left(P_i^2 + Q_i^2\right)}. \tag{22}$$

The load block calculates the specified $R - L$ needed to consume the input signals P_i and Q_i at the rated bus voltage V_{rb} and grid frequency f_{grid}, with a power factor close to unity. This is a typical requirement in data center design for demand response participation [21]. The chilled water and condenser water pump power consumption are modeled as a chain of converters, which represents a simplified average model for the motor drive. It consists of an AC-DC converter, a stage duty cycle DC-DC converter, and a rotational electromechanical converter. The pump power is expressed as;

$$P_{pump} = d\left(V_{ref} - DI_{DC}\right) I_m. \tag{23}$$

where d is the duty cycle of the DC-DC converter, V_{ref} is the DC voltage reference input, used to track the DC bus voltage, and I_m is the current flowing through the motor. The detailed derivation of (22) and (23) are provided in Appendix B and C.

B. Fault Ride-Through Requirements

With the growing integration of AI workload data centers into the grid, ensuring FRT capability has become essential. Without adequate FRT support, and with the rising share of renewable energy sources, data centers may exacerbate transmission system disturbances, which can trigger cascading outages that could lead to large-scale system failures [48]. In response to these challenges, AESO now mandates that the design of any transmission-connected data center must comply with FRT requirements. Specifically, Low- and High-Voltage Ride-Through and Low- and High-Frequency Ride-Through capabilities are required to ensure that data centers remain connected during voltage sags, overvoltage events, and frequency deviations as shown in Table V [12]. This ensures that data centers remain connected during voltage sags and frequency deviations, thereby contributing to grid stability and reducing the risk of widespread disconnections.

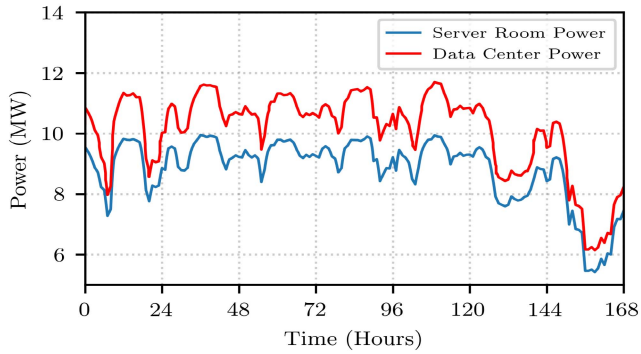


Fig. 12. Total power consumption of the data center.

C. Demand Response

Data centers are considered key players when it comes to demand response participation, aiming to enhance grid stability and efficiency. These facilities can adjust their energy consumption based on signals received from system operators to support grid operations. This work considers the participation of the data center in a negative demand response program by reducing its load consumption.

Algorithm 1 outlines the control strategy, which is used to adjust both the electrical and thermal flexibility of the data center. If a demand response request P_{DR} is received, the server utilization is reduced in proportion to the requested power drop using the defined responsiveness coefficient k_{DR}. This staged reduction ensures that the server load contribution is curtailed without exceeding operational constraints, as the utilization is clamped between 0 and 1. After a predefined delay period, which results in a temperature drop in the server room, the cooling system contributes to additional load reduction by adjusting the chiller supply air temperature. The setpoint $T_{setpoint,t}$ is increased relative to the chiller's share of power reduction, scaled by a temperature factor γ, and bounded within allowable limits in accordance with ASHRAE standards. To ensure practicality, it is essential to account for time delays within the system framework. Therefore, the combined time delay resulting from communication latency and actuation delay is set at 6 ms [49].

VI. STABILITY RESULTS AND DISCUSSION

In this section, the results presented in Fig. 12– 15 represents scenario 1 dynamic performance of the data center over a week with server utilization, while Figs. 16 and 17 correspond to scenario 2, which shows the impact of environmental conditions. Figs. 18 and 19 represent dynamic response under different grid disturbances at 100% server utilization with an average environment temperature of 30 °C and humidity of 60%. Fig. 20 represents the FRT performance, and Figs. 21 and 22 shows the data center response during the demand response event and grid impact, respectively.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

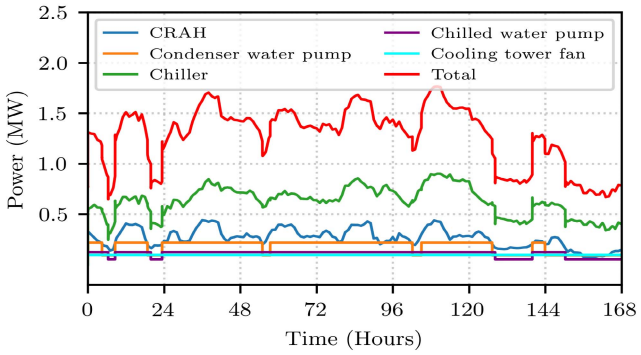


Fig. 13. HVAC system power consumption.

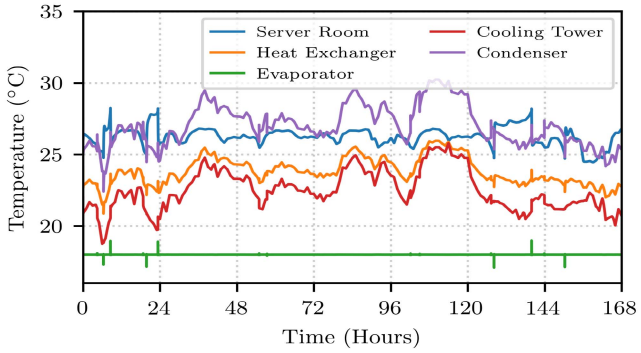


Fig. 14. Data center temperature dynamics.

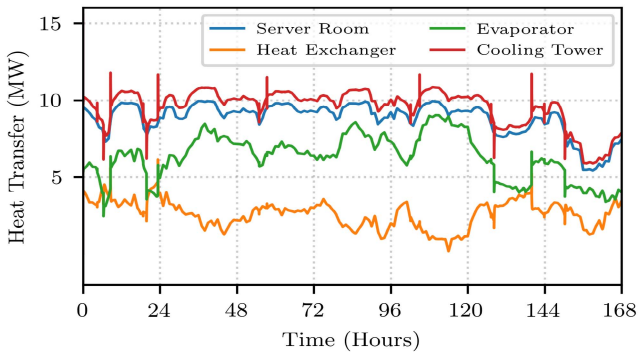


Fig. 15. Data center heat load.

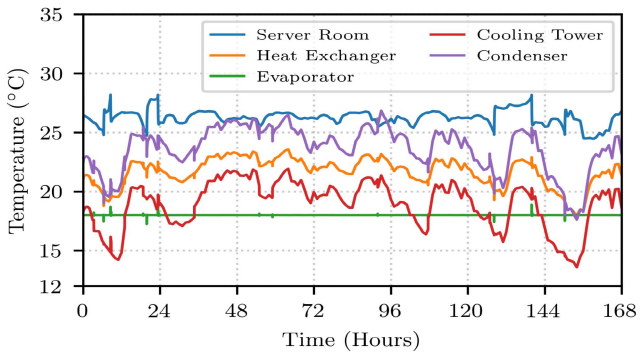


Fig. 16. Data center temperature dynamics for cold environment.

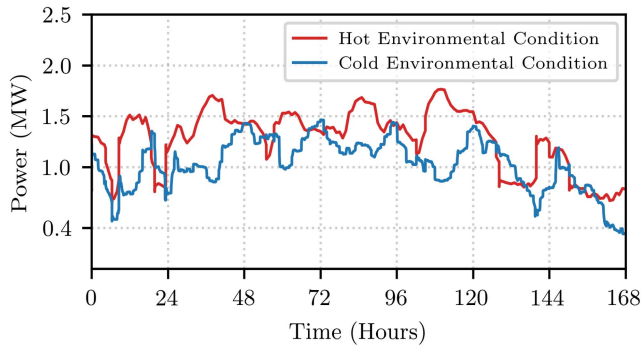


Fig. 17. Environmental impact on HVAC power consumption.

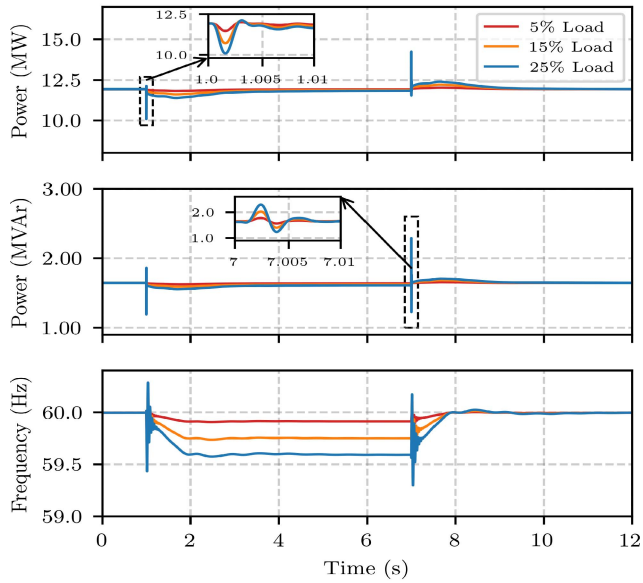


Fig. 18. Load disturbance on data center power and frequency response.

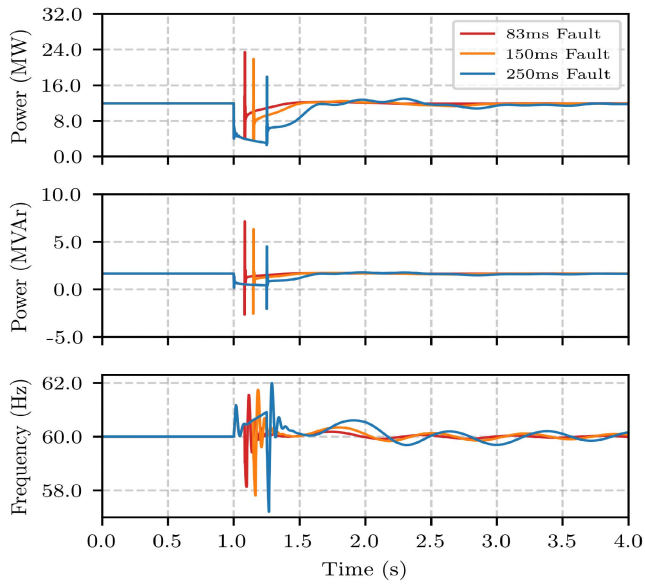


Fig. 19. Fault disturbance on data center power and frequency response.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

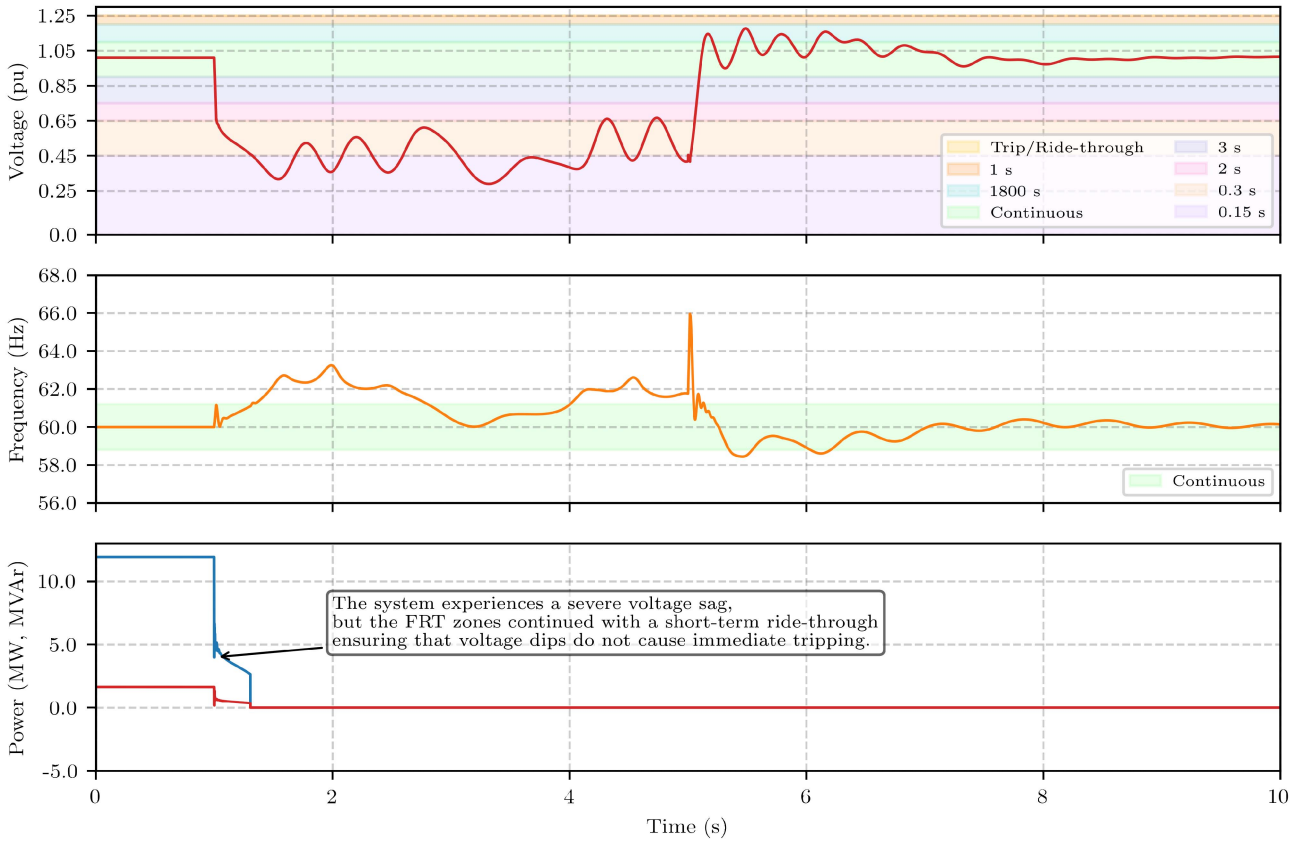


Fig. 20. Data Center Voltage and Frequency Response with FRT Performance.

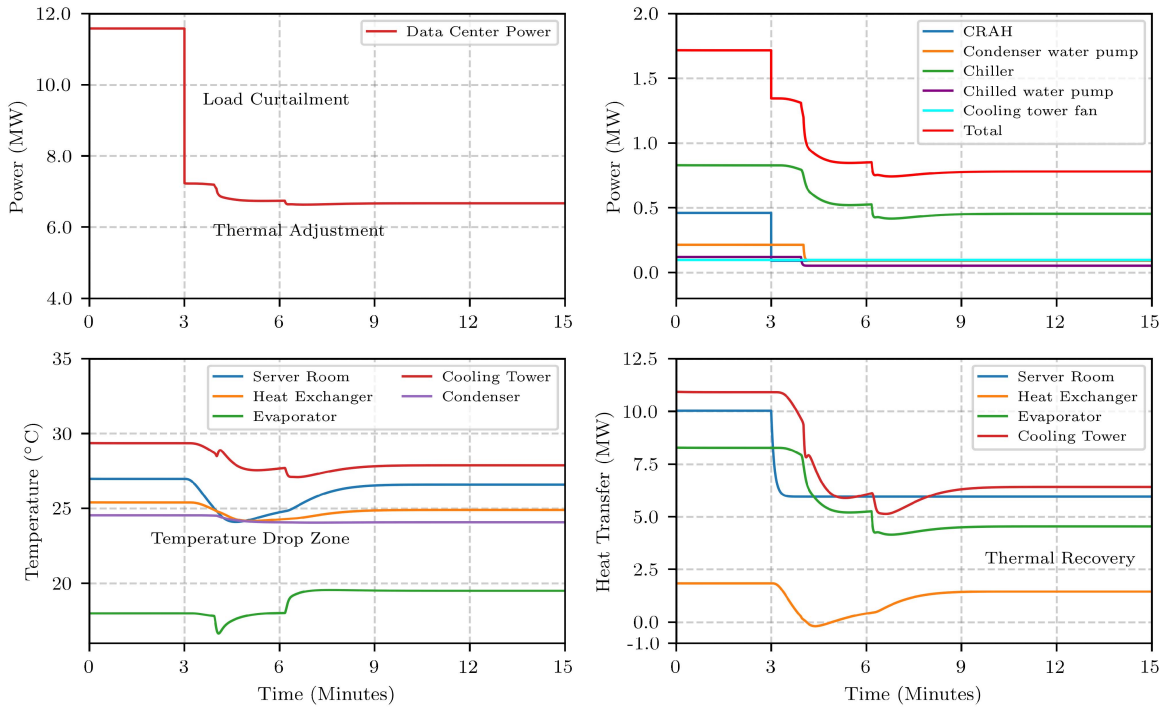


Fig. 21. Data Center response during demand response participation.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: Demand Response Control.

Inputs:

P_{DR}: *requested demand response power reduction;* U_{base}: *baseline server utilization;* k_{DR}: *demand response defined responsiveness coefficient;* $P_{HVAC,t}$: *HVAC power;* $t_{current}$: *simulation time;* t_{DR_start}: *demand response participation start time;* γ: *Temperature scaling factor;* $P_{DR_chiller,t}$: *Chiller Power Drop;*

Outputs:

U_{target}: *Adjusted server utilization;* $T_{setpoint}$: *Adjusted chiller supply air temperature;*

Parameters:

$T_{nom} = 18\,^{\circ}C$ *Nominal chiller supply temperature;* $T_{min} = 15^{\circ}C$ *Minimum allowable setpoint;* $T_{max} = 23^{\circ}C$ *Maximum allowable setpoint;* $delay = 3$ *(Hold time for further chiller adjustment min)*

Procedure:

$U_{target} = U_{base}$

$T_{setpoint} = T_{nom}$

if $P_{DR} > 0$ **then**

  $U_{target,t} = U_{base,t} \times (1 - k_{DR} \cdot \frac{P_{DR,t}}{P_{DC,t}})$

  if $(t - t_{DR_start}) \geq delay$ **then**

    $T_{setpoint,t} = T_{nom} + \gamma \cdot \frac{P_{DR_chiller,t}}{P_{HVAC,t}}$

    $T_{\min} \leq T_{setpoint} \leq T_{\max}$ *:Clamp ASHRAE limits*

  else

    $T_{setpoint,t} = T_{nom}$ *:Hold nominal temperature*

  end if

end if

$0 \leq U_{target} \leq 1$ *:Clamp server utilization*

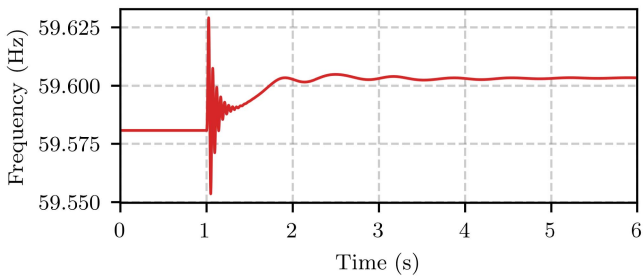


Fig. 22. Frequency dynamics observed under the demand response event.

A. Dynamic Power Consumption and Thermal Load

Fig. 12 illustrates the total power consumption profile for 168 hours. As observed, the power consumption exhibits similar cyclic variations corresponding to the workload in Fig. 7. At peak load, the data center consumes 11.69 MW, with the server room accounting for 9.93 MW (85%) and the remaining 15% attributed to the HVAC system. Accurate monitoring of this power breakdown, especially at peak load, is essential for demand-responsive data centers, particularly in scenarios where renewable energy generation is intermittent or the grid is under stress.

To further demonstrate the effectiveness of the dynamic data center power consumption model, Figs. 13–15 illustrate its ability to simulate dynamic power consumption, temperature dynamics, and heat load of the server room and the HVAC system.

Fig. 13 illustrates the power usage of the data center cooling system, which includes energy contributions from the water pumps (19.3%) and the cooling tower fan (5.4%). However, the main drivers of power consumption are the chiller compressor (50.8%) and the CRAH (24.5%), with a high workload. The total peak power usage, shown in red, reaches 1.77 MW. Therefore, these results suggest that reducing the chiller load could significantly impact total power consumption in a case where the data center is to participate in a demand response event. In addition, the responsiveness of the condenser water pump to load changes will equally optimize power usage.

The temperature dynamics shown in Fig. 14 illustrate the water outlet profile across key stages of the cooling cycle. The chilled water exits the chiller evaporator at a stable 18 °C (green), then rises to approximately 27 °C after absorbing heat in the server room (blue), and reaches around 30.1 °C at the condenser outlet before entering the cooling tower (red). The cooling tower and chiller then work together to bring the temperature back down to 18 °C. These variations directly impact the power consumption of the cooling system, as more significant temperature differentials across the evaporator indicate higher cooling demand, which results in increased chiller load and fan power usage. Understanding these temperature dynamics during demand response events will assist in adjusting and optimizing the cooling system while maintaining a proper temperature for the server room.

Finally, for scenario 1, Fig. 15 presents the heat transfer for various system components. The blue line represents the server room's heat load, which peaks around 9.93 MW. Meanwhile, the red line shows the total heat rejected by the cooling tower, which is slightly higher due to the extra heat generated by the chiller. The upward spikes observed in the cooling tower and evaporator represent critical thermal recovery events that may limit flexibility during demand response periods, which can be interpreted as lockout zones. These effects reflect HVAC compensation for undercooling at high workload, and the graph can help identify such sensitive periods, supporting demand response strategies that factor in thermal safety.

To observe the impact of environmental conditions on the cooling system, we considered a scenario were the data center is located in a cold region with an average temperature and humidity of 22 °C and 70%, respectively. Fig. 16 illustrates the temperature dynamics in the cold environment, while Fig. 17 depicts the HVAC power consumption under hot (scenario 1) and cold environmental conditions. It is worth noting that the system consumes more power in hot environments due to higher cooling demands, while colder conditions lead to reduced power usage as the cooling load decreases.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

B. Dynamic Response Under Different Conditions

The results in Fig. 18 illustrate the dynamic response of the data center model to grid load disturbances. These graphs highlight the impact of the varying load disturbance in 5%, 15%, and 25% of the total system load (315 MW) on the active power, reactive power, and system frequency of the data center. The load is switched on at t = 1 s and off at t = 7 s on bus 6. The active power response reveals an immediate drop followed by transient fluctuations, with a more significant load disturbance inducing more pronounced deviations and oscillatory behavior before regaining stability. Conversely, when the load is disconnected, a similar but opposite pattern is observed, and the magnitude of the power rise is proportional to the level of the disturbance, with the blue line representing 25% load disturbance. Similarly to active power, reactive power exhibits immediate swings followed by damped oscillations, with higher disturbance levels causing greater reactive power fluctuations. The frequency response is particularly critical as it demonstrates a significant drop that corresponds to disturbance conditions. As expected, a higher disturbance (25%) results in a frequency nadir near 59.56 Hz, posing potential risks to the stability of the network and data center. The plots indicate that, despite these transient effects, the system gradually recovers.

In the second scenario, a three-phase fault is applied on line 5-7, as shown in Fig. 10, at t = 1 s after the system reaches a steady state. The results shown in Fig. 19 illustrate the active power, reactive power, and frequency dynamics response to fault-induced disturbance of 83 ms, 150 ms, and 250 ms. Following the fault, the active and reactive power reveals a sharp transient spike with a rapid overshoot when the fault is cleared before regaining stability at nominal levels.

Generally, the stability analysis shows that the system responded successfully to this disturbance and verifies the FRT capabilities of the proposed data center model. The results of this analysis provide important insights when deploying this model in power system dynamic studies. Therefore, it is imperative to deploy detailed data center models in power system dynamics and stability studies to represent the effect of disturbances on the data center load.

C. Fault Ride-Through

Fig. 20 illustrates the voltage, frequency, and power response of the data center at the PCC when the grid is subjected to a 4 s permanent fault. During the initial disturbance, the data center remained connected within the mandatory FRT operation zones. However, after experiencing a voltage sag of 0.45 (pu), it stayed connected for 0.3 s. This indicates that the system effectively withstood the low voltage without immediate disconnection, thereby complying with the FRT requirements specified in Table V. As the sag extended beyond the tolerance range, the data center eventually got disconnected from the grid and is assumed to have switched to on-site backup generation to maintain continuous operation. These results highlight the resilience of data centers under severe grid disturbance. By being able to ride through short-term disturbances, they ensure grid support and system stability, which is particularly critical for transmission-connected data centers participating in demand response.

D. Demand Response

In a scenario where the grid experiences 25% load increase, the data center receives a signal to participate in a demand response event designed to enhance overall grid stability. In response to this, the facility reduces its power consumption by delaying non-urgent tasks for few minutes. The impact of this strategy on the data center's operation is illustrated in Fig. 21. The data center power consumption shows that upon receiving the signal, the facility implemented a staged reduction strategy, decreasing its load to 42.9% of the baseline consumption. The temperature dynamics and heat transfer illustrate the thermal variations within the cooling loop. During the event, the server room temperature gradually decreases to 24.1 °C. However, to further reduce the power consumption of the cooling system, the evaporator temperature is increased, causing the server room temperature to rise to 26.57 °C, which is still within ASHRAE limits. These results demonstrate the necessity for understanding how each component responds in terms of power consumption and thermal flexibility during demand response participation.

The demand response impact on the grid is evident through the frequency dynamics shown in Fig. 22. The system frequency increased from 59.58 Hz to 59.60 Hz due to a sudden reduction in demand. This response shows the data center's ability to provide frequency support and improve grid stability during demand response participation.

VII. Conclusion

In this work, we proposed a comprehensive dynamic model of a data center to depict its dynamic performance under system disturbances. To perform stability studies, we integrated the model into the WSCC 9 bus test bench to investigate the impact of grid disturbance on the data center.

The entire data center model employs a tandem hybrid HVAC cooling approach and includes detailed modeling and validation of server power consumption, heat generation, and temperature dynamics, all of which have been validated against experimental data. The simulation is carried out using the workload of a real-world data center, showing that the proposed model enabled the prediction of the data center's power consumption, heat load, and dynamic temperature performance. These predicted results can aid in monitoring and developing effective energy management strategies, especially when they participate in demand response. Additionally, this study conducted a small signal stability analysis for the data center prior to grid integration to study its impact on power system stability under various disturbances, such as load changes and three-phase short-circuit faults. The results show that, although the data center is affected by the disturbances, the system remains stable, with both active power, reactive power, and frequency returning to acceptable operational conditions. The findings from the stability analysis provide valuable insights into the data center's response to grid disturbances under FRT requirements, as well as its capability to participate in demand response programs for enhancing power

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

system stability. As part of future work, large-scale data center models will be needed, enabling their participation in a broader range of demand response programs. This will be beneficial for frequency regulation and ancillary services provision in renewable energy-dominated power systems.

APPENDIX A
HVAC SYSTEM SIZING

1) *Cooling tower volumetric flow rate:* With water being the main coolant for the condenser water loop, we can use the equation below to calculate the nominal volumetric flow rate of the cooling tower.

$$\dot{V}_{nom} = \frac{\dot{Q}_{thermal}}{\rho \cdot c_p \cdot \Delta T} \tag{A.1}$$

The cooling efficiency of the tower at a temperature difference of $5.2\,^{\circ}\mathrm{C}$ is approximately 0.86, which corresponds to the rule-of-thumb of sizing the tower flow rate at 3 gpm per ton. Here, $\dot{Q}_{thermal}$ is the heat load of the data center, ρ is the density of water (kg/m ^{3}), c_p is the specific heat capacity of water (kJ/kg·$^{\circ}$C), and ΔT is the temperature rise of water across the condenser water loop.

2) *Cooling tower fan power calculation:* To determine the power required for the airflow in the cooling tower fan, we use:

$$P_{fan} = \frac{\dot{V}_a \cdot \Delta P}{\eta_e} \tag{A.2}$$

where P_{fan} is the power required by the fan (W), $\dot{V}_a$ is the airflow rate (m^3/s), ΔP is the pressure increase across the fan (Pa), and η_e is the fan efficiency. In this study, a static fan efficiency of 0.65, a motor efficiency of 0.88, and a fan pressure rise of 190 Pa are considered in calculating the required fan power.

3) *Chiller cooling load:* The cooling load handled by the chiller in both the condenser and chilled water loop, $\dot{Q}_{chiller}$, is given by:

$$\dot{Q}_{chiller} = C_{p,w} \cdot \rho_w \cdot \Delta T_{loop} \cdot \dot{V}_{loop} \tag{A.3}$$

where ΔT_{loop} is the design temperature difference in the cooling loop (K), and $\dot{V}_{loop}$ is the design volumetric flow rate of the cooling loop (m^3/s). The chiller power requirement, $P_{chiller}$, can be determined using the chiller's Coefficient of Performance (COP).

$$P_{chiller} = \frac{\dot{Q}_{chiller}}{COP} \tag{A.4}$$

4) *Heat Exchanger:* To speed up the simulation, a system-level heat exchanger (TL-TL) block is used, which relies on performance data rather than detailed geometry. The block transfers a heat rate $\dot{Q}$ between the chilled and condenser water loops when the inlet temperatures are 28 °C and 22 °C, respectively. The heat balance in the heat exchanger can be expressed as:

$$\dot{Q}_{chilled} + \dot{Q}_{condenser} = 0 \tag{A.5}$$

where $\dot{Q}_{chilled}$ is the heat flow rate from the heat transfer surface wall to the chilled water loop(W), and $\dot{Q}_{condenser}$ is the heat flow rate from the wall to the condenser water loop.

5) *Pump Power Requirement:* The pump power, P_{pump} for both condense and chilled water loop is calculated as;

$$P_{pump} = \frac{\dot{V}_{loop}}{\eta_p} \cdot \Delta P_{loop} \tag{A.6}$$

where $\dot{V}_{loop}$ reflects the loop-specific volumetric flow rate, ΔP_{loop} is the pressure difference across the pump, and η_p is the efficiency of the pump.

6) *CRAH:* This unit regulates chilled water from the chiller, draws warm air from the server room, and recirculates cooled air. Its efficiency depends on the heat exchanger's transfer efficiency and airflow rate. According to [18], the CRAH heat removal process is mathematically modeled using a modified version of the effectiveness-NTU method, as described in (A.7).

$$Q_{crah} = E \cdot \kappa \cdot \dot{m}_{air} \cdot c_{p,loop} \cdot f^{0.7} \left(\kappa T_{ah} + (1-\kappa) T_{ac} - T_{wc}\right) \tag{A.7}$$

Here, E represents heat transfer efficiency ranging from 0 to 1. The containment index κ measures the proportion of air drawn directly from the cold aisle, with $\kappa = 1$ signifying ideal containment and no recirculation of hot air. f is the air volume flow rate expressed as a fraction of the maximum possible volume flow rate. The cooling process is further influenced by the temperature established by T_{ah} (air temperature entering from the hot aisle), T_{ac} (air temperature exiting the cold aisle), and T_{wc} (chilled water temperature). The power consumption of the CRAH in (A.8) is mainly driven by its fan power, which increases significantly with the cube of the mass flow rate, reaching a maximum operational value ($P_{CRAHDyn}$). Furthermore, idle power consumption by the CRAH's control systems ($P_{CRAHIdle}$) contributes to its overall energy usage.

$$P_{CRAH} = P_{CRAH_Idle} + P_{CRAH_Dyn} f^3 \tag{A.8}$$

Assuming total heat rejection by the CRAH due to perfect containment, (A.7) can be used to determine the factor f with the required airflow for a CRAH's maximum standard airflow of 14,000 CMH for a unit with a cooling capacity of 7.5 kW [50].

APPENDIX B
DERIVATION OF EQUIVALENT R AND L PARAMETERS

Starting from a per-phase series impedance $Z = R + jX$, with $X = \omega L$, the per-phase complex power is

$$s = p + jq = \frac{V_{ph}^2}{R + jX}. \tag{B.1}$$

Equating real and imaginary parts gives

$$p = \frac{V_{ph}^2 R}{R^2 + X^2}, \qquad q = \frac{V_{ph}^2 X}{R^2 + X^2}. \tag{B.2}$$

By solving for R, X and L, we obtain

$$R = \frac{p\, V_{ph}^2}{p^2 + q^2}, \qquad X = \frac{q\, V_{ph}^2}{p^2 + q^2}, \qquad L = \frac{X}{\omega}. \tag{B.3}$$

Substituting $p = P/3$, $q = Q/3$, and $V_{ph} = V_{rb}/\sqrt{3}$ yields the compact three-phase form:

$$R = \frac{P\, V_{rb}^2}{P^2 + Q^2}, \qquad L = \frac{Q\, V_{rb}^2}{2\pi f_{grid} \left(P^2 + Q^2\right)}. \tag{B.4}$$

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

This ensures equivalence between the specified load powers and the electrical model.

APPENDIX C
DERIVATION FOR PUMP POWER

The derivation of the pump electrical consumption begins with the AC–DC conversion stage, where the DC bus voltage is regulated as

$$V_{DC} = V_{ref} - DI_{DC}, \tag{C.1}$$

with V_{ref} denoting the reference DC voltage and D the voltage droop parameter. In the DC–DC conversion stage, the motor terminal voltage is related to the duty cycle d as

$$V_m = dV_{DC}. \tag{C.2}$$

For electromechanical conversion, the motor torque and voltage satisfy

$$T = KI_m, \qquad V_m = K\omega, \tag{C.3}$$

where I_m is the motor current, ω the pump speed, and K the motor constant. Assuming negligible losses, the electrical and mechanical powers are equal, i.e.,

$$P_{pump} = V_m I_m = T\omega. \tag{C.4}$$

By substituting (C.2) into (C.4), the pump power becomes

$$P_{pump} = dV_{DC} I_m. \tag{C.5}$$

Finally, using the DC bus expression from (C.1), the pump power consumption is expressed as

$$P_{pump} = d\left(V_{ref} - DI_{DC}\right) I_m. \tag{C.6}$$

REFERENCES

[1] IEA, "What the data centre and ai boom could mean for the energy sector," IEA, Paris, France. 2024. [Online]. Available: https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector

[2] GreenMatch, "Data centre fuels concern for sustainable data solutions," 2024. Published: Sep. 3, 2024. [Online]. Available: https://www.greenmatch.co.uk/blog/data-centre-cooling

[3] M. Gooding, "Global data center electricity use to double by 2026 - IEA Report," 2024. Accessed: Mar. 09, 2024. [Online]. Available: https://www.datacenterdynamics.com/en/news/global-data-center-electricity-use-to-double-by-2026-report/

[4] Data Centre Magazine, "Energy efficiency predictions for data centers in 2023," 2022. Published: Dec. 30, 2022. [Online]. Available: https://datacentremagazine.com/articles/efficiency-to-loom-large-for-data-centre-industry-in-2023

[5] J. Aljbour, T. Wilson, and P. Patel, "Powering intelligence: Analyzing artificial intelligence and data center energy consumption," EPRI White Paper 3002028905, 2024. Accessed: Mar. 09, 2024. [Online]. Available: https://www.epri.com/research/products/000000003002028905

[6] V. Dvorkin, "Agent coordination via contextual regression (Agentconcur) for data center flexibility," *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1832–1842, Mar. 2025.

[7] C. Bash, J. Bian, D. Milojicic, C. D. Patel, L. Strezoski, and V. Terzija, "Energy supplies for future data centers," *Comp.*, vol. 57, no. 7, pp. 126–134, 2024.

[8] X. Long, Y. Li, Y. Li, Y. Liu, P. Yang, and Z. Zeng, "Flexible load scheduling of hydrogen energy data centers: An efficiency-preferred non-convex uncertain optimization approach," *IEEE Trans. Power Syst.*, early access, Aug. 22, 2025, doi: 10.1109/TPWRS.2025.3600683.

[9] Y. Zhang, H. Tang, H. Li, and S. Wang, "Unlocking the flexibilities of data centers for smart grid services: Optimal dispatch and design of energy storage systems under progressive loading," *Energy*, vol. 316, 2025, Art. no. 134511.

[10] I. Drovtar, M. Leinakse, K. Tuttelberg, and J. Kilter, "Utilizing demand response in load modelling for voltage and reactive power control studies," *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1389–1400, Mar. 2025.

[11] C. Jin, X. Bai, C. Yang, W. Mao, and X. Xu, "A review of power consumption models of servers in data centers," *Appl. Energy*, vol. 265, 2020, Art. no. 114806.

[12] Alberta Electric System Operator (AESO), "Aeso connection requirements for transmission-connected data centres: Draft for stakeholder review," draft document for stakeholder review, 2024. [Online]. Available: https://www.aeso.ca/

[13] H. Cheung, S. Wang, C. Zhuang, and J. Gu, "A simplified power consumption model of information technology (IT) equipment in data centers for energy system real-time dynamic simulation," *Appl. Energy*, vol. 222, pp. 329–342, 2018.

[14] S.-W. Ham, M.-H. Kim, B.-N. Choi, and J.-W. Jeong, "Simplified server model to simulate data center cooling energy consumption," *Energy Buildings*, vol. 86, pp. 328–339, 2015.

[15] A. Beghi, L. Cecchinato, G. Dalla Mana, M. Lionello, M. Rampazzo, and E. Sisti, "Modelling and control of a free cooling system for data centers," *Energy Procedia*, vol. 140, pp. 447–457, 2017.

[16] Y. Fu, W. Zuo, M. Wetter, J. W. VanGilder, and P. Yang, "Equation-based object-oriented modeling and simulation of data center cooling systems," *Energy Buildings*, vol. 198, pp. 503–519, 2019.

[17] G. Zhabelova, M. Vesterlund, S. Eschmann, Y. Berezovskaya, V. Vyatkin, and D. Flieller, "A comprehensive model of data center: From CPU to cooling tower," *IEEE Access*, vol. 6, pp. 61254–61266, 2018.

[18] K. M. U. Ahmed, J. Sutaria, M. H. Bollen, and S. K. Rönnberg, "Electrical energy consumption model of internal components in data centers," in *Proc. 2019 IEEE PES Innov. Smart Grid Technol. Europe*, 2019, pp. 1–5.

[19] H. An and X. Ma, "Dynamic coupling real-time energy consumption modeling for data centers," *Energy Rep.*, vol. 8, pp. 1184–1192, 2022.

[20] J. Sun, S. Wang, J. Wang, and L. M. Tolbert, "Dynamic model and converter-based emulator of a data center power distribution system," *IEEE Trans. Power Electron*, vol. 37, no. 7, pp. 8420–8432, Jul. 2022.

[21] A. Yavarian, G. Zhabelova, and V. Vyatkin, "Data center power dynamics within the settings of regional power grid," in *Proc. 20th IEEE Int. Conf. Emerg. Tech. Factory Automat.*, 2015, pp. 1–5.

[22] J. Sun, M. Xu, M. Cespedes, and M. Kauffman, "Low-frequency input impedance modeling of single-phase PFC converters for data center power system stability studies," in *Proc. 2019 IEEE Energy Conv. Congr. Expo.*, 2019, pp. 97–106.

[23] J. Sun, M. Xu, M. Cespedes, D. Wong, and M. Kauffman, "Modeling and analysis of data center power system stability by impedance methods," in *Proc. 2019 IEEE Energy Conv. Congr. Expo.*, 2019, pp. 107–116.

[24] P. Luo, X. Wang, and Y. Li, "The impact of datacenter load regulation on the stability of integrated power systems," *Sust. Energy Tech. Assessments*, vol. 42, 2020, Art. no. 100875.

[25] D. Al Kez, A. M. Foley, S. Muyeen, and D. J. Morrow, "Manipulation of static and dynamic data center power responses to support grid operations," *IEEE Access*, vol. 8, pp. 182078–182091, 2020.

[26] H. Suryanarayana, L. Qi, Y. Zhang, T. Jiang, S. Colombi, and H. Handlin, "System modeling and fault studies in data center power distribution," in *Proc. 26th Int. Conf. Exhib. Electricity Distrib.*, 2021, pp. 1435–1439.

[27] A. Jimenez-Ruiz and F. Milano, "Data center model for transient stability analysis of power systems," 2025, *arXiv:2505.16575*.

[28] ASHRAE Technical Committee 9.9, "Thermal guidelines for data processing environments, 5th Ed.," ASHRAE, 2014. [Online]. Available: https://www.ashrae.org

[29] D. Jiang, "Effects and optimization of airflow on the thermal environment in a data center," *Front. Built Environ.*, vol. 10, 2024, Art. no. 1362861.

[30] J. Chu and X. Huang, "Research status and development trends of evaporative cooling air-conditioning technology in data centers," *Energy Built Environ.*, vol. 4, no. 1, pp. 86–110, 2023.

[31] The MathWorks, Inc., *MATLAB\Simscape, Version 24.2.0.2871072 (R2024b), Update 5*. Natick, MA, USA, 2024. Accessed: Nov. 23, 2024. [Online]. Available: https://www.mathworks.com

[32] M. Dayarathna, Y. Wen, and R. Fan, "Data center energy consumption modeling: A survey," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 732–794, Firstquarter 2016.

[33] W. Wu, L. Jin, J. Yang, P. Liu, and S. X.-D. Tan, "Efficient power modeling and software thermal sensing for runtime temperature monitoring," *ACM Trans. Des. Automat. Electron. Syst.*, vol. 12, no. 3, pp. 1–29, 2008.

[34] X. Fan, W.-D. Weber, and L. A. Barroso, "Power provisioning for a warehouse-sized computer," *ACM SIGARCH Comput. Archit. News*, vol. 35, no. 2, pp. 13–23, 2007.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.

[35] W. Lin, W. Wang, W. Wu, X. Pang, B. Liu, and Y. Zhang, "A heuristic task scheduling algorithm based on server power efficiency model in cloud environments," *Sust. Comput.: Inform. Syst.*, vol. 20, pp. 56–65, 2018.
[36] X. Zhang, J.-J. Lu, X. Qin, and X.-N. Zhao, "A high-level energy consumption model for heterogeneous data centers," *Simul. Model. Prac. Theory*, vol. 39, pp. 41–55, 2013.
[37] S. R. Sarangi, "Power and temperature," in *Next-Gen Computer Architecture: Till the End of Silicon*. Mohali, India: White Falcon Publishing, 2017, ch. 11, pp. 607–636.
[38] S. Sampath, "Thermal analysis of high end servers based on development of detail model and experiments," *Mech. Aerosp. Eng. Theses*, Jan. 2012. [Online]. Available: https://mavmatrix.uta.edu/mechaerospace_theses/102
[39] V. Pandiyan, "Development of detailed computational flow model of high end server and validation using experimental methods," *Mech. Aerosp. Eng. Theses*, Jan. 2012. [Online]. Available: https://mavmatrix.uta.edu/mechaerospace_theses/490
[40] Y. Berezovskaya, C.-W. Yang, A. Mousavi, V. Vyatkin, and T. B. Minde, "Modular model of a data centre as a tool for improving its energy efficiency," *IEEE Access*, vol. 8, pp. 46559–46573, 2020.
[41] Z. Wang, C. Bash, N. Tolia, M. Marwah, X. Zhu, and P. Ranganathan, "Optimal fan speed control for thermal management of servers," in *Proc. ASME 2009 InterPACK Conf., 2009 Summer Heat Transfer Conf., 3rd Int. Conf. Energy Sustain.*, San Francisco, CA, USA, Jul. 19–23, 2009, pp. 709–719, doi: 10.1115/InterPACK2009-89074.
[42] A. Mousavi, V. Vyatkin, Y. Berezovskaya, and X. Zhang, "Towards Energy Smart Data Centers: Simulation of Server Room Cooling System," in *Proc. IEEE 20th Conf. On Emerg. Techn. Factory Automat*, 2015, pp. 1–6.
[43] L. Cupelli, T. Schütz, P. Jahangiri, M. Fuchs, A. Monti, and D. Müller, "Data center control strategy for participation in demand response programs," *IEEE Trans. Ind. Inform.*, vol. 14, no. 11, pp. 5087–5099, Nov. 2018.
[44] M. Ibrahim, S. Shrivastava, B. Sammakia, and K. Ghose, "Thermal mass characterization of a server at different fan speeds," in *Proc. 13th Inter-Soc. Conf. Thermal Thermomechanical Phenomena Electron Syst.*, 2012, pp. 457–465.
[45] S. Samsi et al., "The MIT supercloud dataset," in *Proc. 2021 IEEE High Perform. Extreme Comput. Conf.*, 2021, pp. 1–8.
[46] A. Habibi Khalaj, T. Scherer, and S. K. Halgamuge, "Energy, environmental and economical saving potential of data centers with various economizers across Australia," *Appl. Energy*, vol. 183, pp. 1528–1549, 2016.
[47] D. Mondal, A. Chakrabarti, and A. Sengupta, *Power System Small Signal Stability Analysis and Control*. New York, NY, USA: Academic Press, 2020.
[48] T. Kerçi, C. Duggan, U. Farooq, S. Tweed, and M. Val Escudero, "Impact of converter-based demand on frequency quality in the Ireland and northern Ireland power systems," in *Proc. C4 PS1 - Power Syst. Dyn. Anal. Energy Transition: Challenges, Opportunities Adv.*, Paris, France, Aug. 2024, pp. 1–11.
[49] D. Al Kez, A. M. Foley, F. W. Ahmed, M. O'Malley, and S. Muyeen, "Potential of data centers for fast frequency response services in synchronously isolated power systems," *Renewable Sustain. Energy Rev.*, vol. 151, 2021, Art. no. 111547.
[50] R. Rahmani, I. Moser, and M. Seyedmahmoudian, "A complete model for modular simulation of data centre power load," 2018, *arXiv:1804.00703*.



Pam Paul Gyang (Student Member, IEEE) received the B.Eng. degree in electrical and electronics engineering (power systems option) from the Federal University of Technology, Owerri, Nigeria, in 2015, and the M.Sc. degree in electrical and electronics engineering (electrical power option) from the University of Lagos, Akoka, Nigeria, in 2020. He is currently working toward the Cotutelle (Joint-Supervision) Ph.D. degree in electrical engineering with the Birla Institute of Technology and Science, Hyderabad, India, and RMIT University, Melbourne, VIC, Australia. From 2018 to 2023, he was an Engineering Standards and Research Specialist with Eko Electricity Distribution Company, Lagos, Nigeria. His research interests include power system stability and dynamics, demand response and flexible load control, dynamic modeling of grid-interactive data centers, and stability assessment of renewable energy dominated power systems.



Pratyush Chakraborty (Senior Member, IEEE) received the B.E. degree in electrical engineering from Jadavpur University, Kolkata, India, in 2006, the M.Tech. degree in electrical engineering from IIT Bombay, Mumbai, India, in 2011, and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Florida, Gainesville, FL, USA, in 2013 and 2016, respectively. From 2006 to 2009, he worked in Siemens Limited, India. From 2017 to 2020, he was Postdoctoral Researcher with the three Institutes of USA, which are University of California, Berkeley, Northwestern University, and University of Utah. He is currently an Associate Professor with the Department of Electrical and Electronics Engineering, BITS Pilani, Hyderabad, India. His research interests include game theory, optimization, control theory and their applications to power systems with deep renewable penetration.



Lasantha Meegahapola (Senior Member, IEEE) received the Ph.D. degree from Queen's University Belfast, Belfast, U.K., in 2010. He has more than eighteen years of research experience in power system dynamics and stability with renewable power generation and has authored or coauthored more than 230 journal and conference articles. From 2009 to 2010, he was a Visiting Researcher with the Electricity Research Centre, University College Dublin, Dublin, Ireland. From 2011 to 2014, he was a Lecturer with the University of Wollongong (UOW), Wollongong, NSW, Australia, and continues as a Senior Honorary Fellow. He is currently a Professor of electrical power systems with the Royal Melbourne Institute of Technology (RMIT) University, Melbourne, VIC, Australia. He is also an Associate Editor for IEEE TRANSACTIONS ON POWER SYSTEMS, IEEE PES LETTERS, and IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS. He is also a Member of the IEEE Power Engineering Society (PES) and IEEE Industrial Electronics Society (IES). He is also an Active Member of the IEEE PES power system dynamic performance (PSDP) committee task forces on 'Microgrid Stability Definitions, Analysis, and Modeling' and 'Microgrid Dynamic Modeling'.



Xinghuo Yu (Fellow, IEEE) received the B.Eng. and M.Eng. degrees in electrical and electronic engineering from the University of Science and Technology of China, Hefei, China, in 1982 and 1984, respectively, and the Ph.D. degree in control science and engineering from Southeast University, Nanjing, China, in 1988. He is a Distinguished Professor and an Associate Deputy Vice-Chancellor with the Royal Melbourne Institute of Technology (RMIT University), Melbourne, VIC, Australia. His research interests include control systems, complex and intelligent systems, and power and energy systems. He was the recipient of the number of awards and honors for his contributions, including 2013 Dr.-Ing. Eugene Mittelmann Achievement Award of IEEE Industrial Electronics Society and 2018 M. A. Sargent Medal from Engineers Australia. In 2018 and 2019, he was the President of IEEE Industrial Electronics Society. He was an Associate Editor for IEEE TRANSACTIONS ON AUTOMATIC CONTROL, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I: REGULAR PAPERS, IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, and several other journals. He is a Fellow of the Australian Academy of Science, Honorary Fellow of Engineers Australia, and a Fellow of International Federation of Automatic Control.

Authorized licensed use limited to: Xiamen University of Technology. Downloaded on October 15,2026 at 01:41:21 UTC from IEEE Xplore. Restrictions apply.